Endogenous Education Choice in Segmented Search Markets

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1 Global environment

Time is continuous. All agents discount at rate $r > 0$.

Aggregate productivity. Aggregate productivity z_t is piecewise constant and takes values in a finite set $\mathcal{Z} = \{z_L, z_H\}$ (or more generally $\{z_1, \ldots, z_K\}$). A *regime change* occurs at an (exogenous) stopping time T, at which z_t jumps from an initial value z_0 to a new value $z_1 \neq z_0$:

$$z_t = \begin{cases} z_0, & t < T, \\ z_1, & t \geq T, \end{cases}$$

The shock is *unexpected* (agents do not condition decisions on an anticipated switch date) and *permanent* (once z switches, it remains at the new level forever).

Stationarity and transitions. For a fixed regime $z \in \mathcal{Z}$, the economy admits a stationary equilibrium (steady state) in which the cross-sectional distribution of workers across states is time-invariant. A regime change at time T generates a *transition path*: value functions and policies adjust immediately to the new regime z_1, while the cross-sectional distribution of workers is predetermined at T and evolves gradually according to flow balance equations.

Sectoral productivity shifters. Each regime $z \in \mathcal{Z}$ maps into sector-specific productivity shifters $P_U(z) > 0$ and $P_S(z) > 0$. These shifters are constant within a regime and jump permanently at regime changes.

2 Workers, firms, and types

2.1 Workers

Each worker has an innate type $x \in \mathcal{X} = [0, 1]$, fixed over time. Types are distributed according to CDF L with density ℓ ($x \sim \text{Beta}(a_x, b_x)$).

Market access via training (persistent eligibility). Workers can participate in either:

- The *unskilled* market U (untrained status),
- The *skilled* market S (trained status).

Training changes *market access only*; it does not change innate type x.

Training decision at unskilled unemployment. Training can be initiated *only* by untrained workers who are in unskilled unemployment. For each regime z, the optimal policy is characterized by a cutoff $\bar{x}(z) \in [0, 1]$: upon entering unskilled unemployment in regime z, types $x > \bar{x}(z)$ initiate training, while types $x \leq \bar{x}(z)$ remain in unskilled search.

Interpretation under regime changes. Because training status is persistent, a regime change affects only the *flow into training* from the pool of untrained unemployed. After a regime change, the cutoff $\bar{x}(z)$ associated with the new regime governs the training decision of newly-arriving untrained unemployed workers, but the economy inherits the pre-shock stock of trained workers. Hence, after a switch (e.g. $z_L \to z_H$), the fraction trained adjusts gradually due to demographic turnover and equilibrium flows.

2.2 Firms

There are two labor markets. Both markets have homogeneous firms.

3 Training technology

Eligibility. Only untrained workers in unskilled unemployment may initiate training.

Cost. Initiating training requires a one-time cost $c(x)$ with $c'(x) < 0$. In applications, $c(x) = a_c e^{-x}$.

Completion. Training completes at Poisson rate $\phi > 0$. Upon completion, the worker becomes trained ($e = 1$) and enters skilled unemployment.

Flow payoffs. Let $b_U(z)$, $b_T(z)$, and $b_S(z)$ denote flow payoffs in regime z while unskilled unemployed, training, and skilled unemployed, respectively.

Demographic turnover (stationarity device). Workers exit the economy at Poisson rate $\nu > 0$ and are replaced one-for-one by newborn untrained workers with density $\ell(x)$. This delivers a stationary population composition within each fixed regime.

4 Production technologies

Unskilled production. In regime z, an unskilled match between worker type x and a homogeneous firm with match-specific efficiency $p \in (0, 1]$ produces

$$\pi_U(x, p, z) = P_U(z)\, x\, p.$$

Skilled production. In regime z, an unskilled match between worker type x and a homogeneous firm with match-specific efficiency $p \in (0, 1]$ produces

$$\pi_S(x, p, z) = P_S(z)\, x\, p.$$

5 Unskilled labor market

Fix a regime $z \in \mathcal{Z}$.

5.1 Matching

Let $\mathcal{U}_U(z)$ and $\mathcal{V}_U(z)$ denote the masses of unskilled unemployed and unskilled vacancies in regime z. Market tightness is

$$\theta_U(z) = \frac{\mathcal{V}_U(z)}{\mathcal{U}_U(z)}.$$

Matching follows

$$M_U(\mathcal{U}_U(z), \mathcal{V}_U(z)) = \mu_U \, \mathcal{U}_U(z)^{\eta_U} \mathcal{V}_U(z)^{1-\eta_U}, \qquad \eta_U \in (0,1),$$

implying vacancy filling and job-finding rates

$$q_U(\theta_U(z)) = \mu_U \, \theta_U(z)^{-\eta_U}, \qquad \theta_U(z) \, q_U(\theta_U(z)).$$

5.2 Match productivity and shocks

New matches are created at frontier efficiency $p_0 = 1$.

Idiosyncratic damage shocks arrive at Poisson rate $\lambda_U > 0$. Upon a shock, post-shock efficiency is

$$p' = e^{-d}, \qquad d \mid z \sim \mathrm{Exp}\big(\alpha(z)\big),$$

equivalently

$$p' \mid z \sim \mathrm{Beta}\big(\alpha(z), 1\big), \qquad G_z(p) = p^{\alpha(z)}, \quad p \in [0,1].$$

A match is endogenously destroyed whenever $p' < p^*(x,z)$ for a reservation threshold $p^*(x,z) \in (0,1)$.

5.3 Value functions and training policy

Let

- $U_U(x,z)$ be the value of untrained unskilled unemployment,
- $U_U^{\text{search}}(x,z)$ the value of unskilled search,
- $T(x,z)$ the value of training,
- $E_U(x,p,z)$ the worker value of unskilled employment at efficiency p,
- $J_U(x,p,z)$ the firm value of a filled unskilled job,
- $V_U(z)$ the value of an unskilled vacancy.

Household values and training policy. Let $U_U(x,z)$ denote the value of *untrained* unskilled unemployment for type x in regime z. An untrained unemployed worker chooses between unskilled search and initiating training:

$$U_U(x,z) = \max\big\{U_U^{\text{search}}(x,z),\ U_U^{\text{train}}(x,z)\big\}, \qquad U_U^{\text{train}}(x,z) := -c(x) + T(x,z).$$

Training completes at Poisson rate ϕ and leads to skilled unemployment. The training value satisfies

$$(r + \phi + \nu)\, T(x,z) = b_T(z) + \phi \, U_S(x,z). \tag{1}$$

Unskilled search yields flow payoff $b_U(z)$ and contacts vacancies at rate $\theta_U(z) q_U(\theta_U(z))$:

$$(r + \nu)\, U_U^{\text{search}}(x,z) = b_U(z) + \theta_U(z) q_U(\theta_U(z)) \big(E_U(x,1,z) - U_U^{\text{search}}(x,z)\big). \tag{2}$$

Define the net training gain

$$\Delta_T(x,z) := -c(x) + T(x,z) - U_U^{\text{search}}(x,z).$$

Under monotone incentives, there exists a cutoff $\bar{x}(z) \in [0,1]$ such that

$$\tau(x,z) = \mathbb{1}\{x > \bar{x}(z)\}, \qquad U_U^{\text{search}}(\bar{x}(z),z) = -c(\bar{x}(z)) + T(\bar{x}(z),z).$$

Match values, bargaining, and endogenous separation A new unskilled match starts at frontier efficiency $p_0 = 1$. Idiosyncratic damage shocks arrive at rate λ_U and redraw efficiency p' from G_z on $[0,1]$.

For $p \in (0,1]$, worker and firm values satisfy

$$(r+\nu+\lambda_U)\, E_U(x,p,z) = w_U(x,p,z) + \lambda_U \left[U_U(x,z)\, G_z\big(p^*(x,z)\big) + \int_{p^*(x,z)}^1 E_U(x,p',z)\ \mathrm{d}G_z(p') \right], \tag{3}$$

$$(r+\nu+\lambda_U)\, J_U(x,p,z) = P_U(z)\, x\, p - w_U(x,p,z) + \lambda_U \int_{p^*(x,z)}^1 J_U(x,p',z)\ \mathrm{d}G_z(p'). \tag{4}$$

Wages are determined by Nash bargaining with worker weight β_U:

$$\beta_U\, J_U(x,p,z) = (1-\beta_U)\big(E_U(x,p,z) - U_U(x,z)\big). \tag{5}$$

Endogenous separation is characterized by a reservation threshold $p^*(x,z) \in (0,1)$ satisfying

$$J_U\big(x,p^*(x,z),z\big) = 0, \qquad J_U(x,p,z) \geq 0 \iff p \geq p^*(x,z). \tag{6}$$

5.4 Vacancies, free entry, and market tightness

Let $V_U(z)$ denote the value of an unskilled vacancy in regime z. Posting a vacancy costs k_U per unit time and yields contacts at rate $q_U(\theta_U(z))$.

Let $u_U(x,z)$ denote the density of untrained unskilled unemployed by type and define

$$\mathcal{U}_U(z) = \int_0^1 u_U(x,z)\ \mathrm{d}x,$$

A vacancy meets type $x \sim u_U(\cdot,z)$ and creates a match at $p_0 = 1$, yielding value $J_U(x,1,z)$. The vacancy HJB is

$$rV_U(z) = -k_U + q_U(\theta_U(z)) \left(\int_0^1 J_U(x,1,z)\, u_U(x,z)\ \mathrm{d}x - V_U(z) \right). \tag{7}$$

Free entry implies

$$V_U(z) = 0 \qquad \Longrightarrow \qquad q_U(\theta_U(z)) = \frac{k_U}{\int_0^1 J_U(x,1,z)\, u_U(x,z)\ \mathrm{d}x}, \tag{8}$$

which pins down market tightness $\theta_U(z)$ given the endogenous composition of unemployed workers.

Implications for Nash bargaining and separation. Nash bargaining (5) implies the standard surplus split

$$E_U(x,p,z) - U_U(x,z) = \beta_U\, S_U(x,p,z), \qquad J_U(x,p,z) = (1-\beta_U)\, S_U(x,p,z).$$

Hence the reservation rule (6) is equivalently

$$J_U\big(x,p^*(x,z),z\big) = 0 \iff S_U\big(x,p^*(x,z),z\big) = 0.$$

5.5 A pointwise update for unskilled search values

Although the training option implies that the bargaining outside value is the full unemployment value

$$U_U(x,z) = \max\{U_U^{\text{search}}(x,z),\, U_U^{\text{train}}(x,z)\},$$

the search value itself admits a convenient pointwise representation.

Rearranging (2) gives

$$\big(r+\nu+\theta_U(z)q_U(\theta_U(z))\big)\, U_U^{\text{search}}(x,z) = b_U(z) + \theta_U(z)q_U(\theta_U(z))\, E_U(x,1,z). \tag{9}$$

Using Nash bargaining at entry efficiency $p_0 = 1$,

$$E_U(x,1,z) = U_U(x,z) + \frac{\beta_U}{1-\beta_U} J_U(x,1,z),$$

we obtain the pointwise update

$$U_U^{\text{search}}(x,z) = \frac{b_U(z) + \theta_U(z)q_U(\theta_U(z))\left(U_U(x,z) + \dfrac{\beta_U}{1-\beta_U} J_U(x,1,z)\right)}{r+\nu+\theta_U(z)q_U(\theta_U(z))}. \tag{10}$$

Under the cutoff policy $\tau(x,z) = \mathbb{1}\{x > \bar{x}(z)\}$,

$$U_U(x,z) = \begin{cases} U_U^{\text{search}}(x,z), & x \le \bar{x}(z), \\ U_U^{\text{train}}(x,z), & x > \bar{x}(z), \end{cases}$$

so (10) becomes a simple piecewise update. In particular, on the search region $x \le \bar{x}(z)$ it reduces to

$$(r+\nu)\, U_U^{\text{search}}(x,z) = b_U(z) + \frac{\beta_U}{1-\beta_U}\, \theta_U(z)q_U(\theta_U(z))\, J_U(x,1,z).$$

This representation is especially convenient for computation: $U_U^{\text{search}}(x,z)$ can be updated pointwise given current $J_U(x,1,z)$ and the cutoff $\bar{x}(z)$, while free entry pins down tightness only through the *composition-weighted* frontier surplus.

We use (10) as the pointwise update for $U_U^{\text{search}}(x,z)$ inside the value-function iteration described in the solution algorithm below.

Unskilled match surplus (wage-free equation). Define the total match surplus as worker net gain plus firm value:

$$S_U(x,p,z) \;\equiv\; \big(E_U(x,p,z) - U_U(x,z)\big) \;+\; J_U(x,p,z).$$

Equivalently, $E_U(x,p,z) + J_U(x,p,z) = S_U(x,p,z) + U_U(x,z)$.

Add the worker and firm HJBs (3)–(4):

$$\begin{aligned}(r+\nu+\lambda_U)\big(E_U(x,p,z) + J_U(x,p,z)\big) &= P_U(z)\, x\, p \\ &\quad + \lambda_U \left[U_U(x,z)\, G_z\big(p^*(x,z)\big) + \int_{p^*(x,z)}^{1} \big(E_U(x,p',z) + J_U(x,p',z)\big)\, \mathrm{d}G_z(p') \right].\end{aligned}$$

Substitute $E_U + J_U = S_U + U_U$ and use

$$\int_{p^*(x,z)}^{1} U_U(x,z)\, \mathrm{d}G_z(p') = U_U(x,z)\big(1 - G_z(p^*(x,z))\big),$$

to simplify the U_U terms:

$$(r+\nu+\lambda_U)\big(S_U(x,p,z)+U_U(x,z)\big) = P_U(z)\,x\,p + \lambda_U\left[U_U(x,z) + \int_{p^*(x,z)}^{1} S_U(x,p',z)\ \mathrm{d}G_z(p')\right].$$

Cancel $\lambda_U U_U(x,z)$ from both sides and rearrange to obtain the wage-free surplus equation:

$$(r+\nu+\lambda_U)\,S_U(x,p,z) = P_U(z)\,x\,p - (r+\nu)\,U_U(x,z) + \lambda_U \int_{p^*(x,z)}^{1} S_U(x,p',z)\ \mathrm{d}G_z(p'). \quad (11)$$

Closed-form wage schedule (unskilled). Given the surplus split $J_U = (1-\beta_U)S_U$ and $E_U - U_U = \beta_U S_U$, wages are recovered directly from the firm HJB (4). Multiplying the surplus equation (11) by $(1-\beta_U)$ and comparing with the firm HJB gives

$$w_U(x,p,z) = \beta_U\,P_U(z)\,x\,p\ +\ (1-\beta_U)\,(r+\nu)\,U_U(x,z). \quad (12)$$

The wage is linear in match quality p: the worker captures share β_U of flow output $P_U(z)\,x\,p$, plus a flow option value $(1-\beta_U)(r+\nu)U_U(x,z)$ reflecting the firm's outside value when the worker exits (at rate ν) or the match is destroyed and the worker returns to U_U. In particular, (12) does not depend on the tail integral $I(x,z) = \int_{p^*}^{1} S_U\ \mathrm{d}G_z$: the shock-option terms cancel exactly between the worker and firm HJBs when wages are eliminated via Nash bargaining.

5.6 Transition dynamics within the unskilled block

We now describe the law of motion of cross-sectional densities in the unskilled block. This serves two purposes: (i) it nests the stationary flow-balance conditions as a special case, and (ii) it provides the transition path after an unexpected, permanent regime change.

Time-dependent objects. Let $z \in \mathcal{Z}$ denote the current regime. For $t \geq 0$, let $u_U(x,t;z)$ be the density of *untrained* workers in unskilled unemployment, and $t(x,t;z)$ the density of workers in training. Define the untrained employed density as the residual

$$e_U(x,t;z) := \ell(x) - u_U(x,t;z) - t(x,t;z),$$

i.e. within the untrained population (mass $\ell(x)$), each type is either unemployed, in training, or employed in an unskilled job.

Policy and hazard terms. Fix a regime z and let the unskilled-block policies and hazards be evaluated at that regime:

$$\tau(x;z) \in \{0,1\}, \qquad \theta_U(z), \qquad p^*(x;z), \qquad \delta_U(x;z) := \lambda_U\,G_z\big(p^*(x;z)\big).$$

Here $\delta_U(x;z)$ is the endogenous job-destruction hazard for an employed untrained type-x worker.

Training stock dynamics. Training receives inflow from untrained unemployment at rate $\tau(x;z)$ and loses mass through completion at rate ϕ and demographic exit at rate ν:

$$\partial_t t(x,t;z) = \tau(x;z)\,u_U(x,t;z) - (\phi+\nu)\,t(x,t;z). \quad (13)$$

Untrained unemployment dynamics. Untrained unemployment receives inflow from demographic replacement and job destruction, and loses mass due to job finding, training entry, and demographic exit. This yields

$$\partial_t u_U(x,t;z) = \nu\,\ell(x) + \delta_U(x;z)\,e_U(x,t;z) - \Big(\theta_U(z)q_U(\theta_U(z)) + \tau(x;z) + \nu\Big)\,u_U(x,t;z). \quad (14)$$

Using $e_U(x,t;z) = \ell(x) - u_U(x,t;z) - t(x,t;z)$, this can be written as a closed system in (u_U, t).

Stationary equilibrium as a special case. Within a fixed regime z, a stationary distribution satisfies $\partial_t u_U(x,t;z) = 0$ and $\partial_t t(x,t;z) = 0$, so (13) and (14) reduce to the density flow-balance equations used above.

Transition after a regime change. Consider an unexpected, permanent switch at calendar time τ from z_0 to z_1. Let the pre-switch stationary densities be $(u_U^0(x), t^0(x))$ (the steady state under z_0). At the instant of the switch, the cross-sectional distribution is predetermined:

$$u_U(x,\tau^+;z_1) = u_U^0(x), \qquad t(x,\tau^+;z_1) = t^0(x).$$

Values and policies adjust immediately to the new regime z_1, implying that $\tau(\cdot;z_1)$, $p^*(\cdot;z_1)$, and $\theta_U(z_1)$ are used in the post-switch dynamics (13)–(14). The transition path is then the solution for $t \mapsto (u_U(\cdot,t;z_1), t(\cdot,t;z_1))$ starting from the initial condition above.

Aggregate masses and flow accounting. Aggregate stocks are obtained by integration:

$$\mathcal{U}_U(t;z) = \int_0^1 u_U(x,t;z)\,\mathrm{d}x, \qquad \mathcal{T}(t;z) = \int_0^1 t(x,t;z)\,\mathrm{d}x, \qquad \mathcal{E}_U(t;z) = \int_0^1 e_U(x,t;z)\,\mathrm{d}x.$$

5.7 Cross-sectional distributions within the unskilled block

Notation. All flow-balance equations in this subsection are written in *density form* with respect to x. Thus $u_U(x,z)$, $t(x,z)$, and $\ell(x)$ are densities, and aggregate masses are obtained by integrating over x.

Within a fixed regime z, the cross-sectional distribution of *untrained* workers across states is pinned down by flow balance. Let $u_U(x,z)$ denote the density of untrained workers in unskilled unemployment and let $t(x,z)$ denote the density of workers in training.

Training. At equilibrium the worker in training distribution is stationary:

$$0 = \tau(x,z)\,u_U(x,z) - (\phi+\nu)\,t(x,z), \qquad \Longrightarrow \qquad t(x,z) = \frac{\tau(x,z)}{\phi+\nu}\,u_U(x,z). \quad (15)$$

Untrained unemployment. Untrained unemployment receives inflow from demographic replacement and job destruction, and loses mass due to job finding, training entry, and demographic exit. Since damage shocks redraw efficiency i.i.d. from G_z, the job-destruction hazard for an employed type-x worker is $\lambda_U\,G_z(p^*(x,z))$. Hence the density flow balance is

$$\Big(\theta_U(z)q_U(\theta_U(z)) + \tau(x,z) + \nu\Big)\,u_U(x,z) = \nu\,\ell(x) + \lambda_U\,G_z\big(p^*(x,z)\big)\,\Big(\ell(x) - u_U(x,z) - t(x,z)\Big). \quad (16)$$

5.8 Equilibrium and solution algorithm

Fix a regime $z \in \mathcal{Z}$. An *unskilled-block equilibrium* in regime z is a collection

$$\Big\{\theta_U(z),\ U_U^{\text{search}}(\cdot,z),\ T(\cdot,z),\ U_U(\cdot,z),\ E_U(\cdot,\cdot,z),\ J_U(\cdot,\cdot,z),\ p^*(\cdot,z),\ u_U(\cdot,z),\ t(\cdot,z)\Big\}$$

such that:

(E1) **Household values and training.** $T(x,z)$ satisfies (1); $U_U^{\text{search}}(x,z)$ satisfies (2) (equivalently (10)); and unemployment values satisfy

$$U_U(x,z) = \max\{U_U^{\text{search}}(x,z),\ -c(x)+T(x,z)\}, \qquad \tau(x,z) = \mathbb{1}\{-c(x)+T(x,z) \geq U_U^{\text{search}}(x,z)\}.$$

(E2) **Match values and wages.** Worker and firm match values satisfy (3)–(4); wages satisfy Nash bargaining (5). (Equivalently, $w_U(x,p,z)$ is recovered from (5) given (E_U, J_U, U_U).)

(E3) **Endogenous separation.** The reservation threshold $p^*(x,z)$ satisfies (6).

(E4) **Vacancy creation (free entry).** Market tightness $\theta_U(z)$ satisfies free entry (8).

(E5) **Stationary cross-sectional distributions.** Densities satisfy the flow-balance conditions (15) and (16). Employment of untrained workers is the residual mass $e_U(x,z) = \ell(x) - u_U(x,z) - t(x,z)$, and the implied contact composition $u_U(\cdot,z)$ enters vacancy creation.

Solution algorithm (unskilled side; fixed regime z). Fix a regime $z \in \mathcal{Z}$. For computation, it is convenient to organize the unskilled block as a two-layer fixed point. The *inner loop* solves the value functions $(U_U^{\text{search}}, U_U, T, E_U, J_U)$ given conjectures for market tightness θ_U and the reservation rule $p^*(\cdot)$. The *outer loop* updates the policy objects $(\tau(\cdot), p^*(\cdot), \theta_U)$ and the stationary composition (u_U, t) until all equilibrium conditions hold jointly.

Outer-loop unknowns. We iterate on

$$\Big(\theta_U^{(n)}(z),\ p^{*,(n)}(\cdot,z),\ \tau^{(n)}(\cdot,z),\ u_U^{(n)}(\cdot,z),\ t^{(n)}(\cdot,z)\Big).$$

Step 1. Initialize (outer loop). Choose initial guesses $\theta_U^{(0)}(z) > 0$ and a feasible reservation rule $p^{*,(0)}(x,z) \in (0,1)$ (e.g. constant in x). Initialize a training policy $\tau^{(0)}(x,z)$ (or cutoff $\bar{x}^{(0)}(z)$) and stationary densities $u_U^{(0)}(x,z)$ and $t^{(0)}(x,z)$ (e.g. proportional to ℓ).

Step 2. Inner value-function iteration (given $\theta_U^{(n)}$ and $p^{*,(n)}$). Holding $(\theta_U^{(n)}(z), p^{*,(n)}(\cdot,z))$ fixed, iterate jointly on

$$\big(U_U^{\text{search}}, U_U, T, E_U, J_U\big)$$

until convergence. One convenient inner-loop cycle is:

Step 2.1. Training value. Update $T^{(m+1)}(x,z)$ from (1):

$$(r+\phi+\nu)\,T^{(m+1)}(x,z) = b_T(z) + \phi\, U_S(x,z),$$

treating $U_S(\cdot,z)$ as given by the skilled block (or fixed when solving the unskilled block in isolation).

Step 2.2. Unemployment value and training decision. Given $T^{(m+1)}$ and the current search value $U_U^{\text{search},(m)}$, update the full outside value:

$$U_U^{(m+1)}(x,z) = \max\Big\{U_U^{\text{search},(m)}(x,z),\ -c(x) + T^{(m+1)}(x,z)\Big\},$$

and the implied training policy

$$\tau^{(m+1)}(x,z) = \mathbb{1}\{-c(x) + T^{(m+1)}(x,z) \geq U_U^{\text{search},(m)}(x,z)\}.$$

Step 2.3. Match values. Given $U_U^{(m+1)}(x,z)$ and the fixed reservation rule $p^{*,(n)}(\cdot,z)$, solve the coupled system (3)–(4) together with bargaining (5) to obtain $E_U^{(m+1)}(x,p,z)$ and $J_U^{(m+1)}(x,p,z)$. (Here feasibility is enforced by integrating only over $p' \geq p^{*,(n)}(x,z)$.)

Step 2.4. Search value. Update $U_U^{\text{search},(m+1)}(x,z)$ from the HJB (2):

$$(r+\nu)\, U_U^{\text{search},(m+1)}(x,z) = b_U(z) + \theta_U^{(n)}(z) q_U(\theta_U^{(n)}(z))\Big(E_U^{(m+1)}(x,1,z) - U_U^{\text{search},(m+1)}(x,z)\Big),$$

Stop the inner loop when the sup norm of updates to $(U_U^{\text{search}}, U_U, E_U, J_U)$ is below tolerance.

Step 3. Update stationary distributions (given the converged policy and hazards). Using the converged training policy $\tau^{(n+1)}(\cdot,z)$ from the inner loop and the current reservation rule $p^{*,(n)}(\cdot,z)$, update the stationary densities $(u_U^{(n+1)}, t^{(n+1)})$ by solving (15)–(16) pointwise in x (with $e_U(x,z) = \ell(x) - u_U(x,z) - t(x,z)$).

Step 4. Update the reservation rule p^* (outer update). Setting $S_U(x,p^*,z) = 0$ in the closed-form surplus expression yields the *direct formula*

$$p^{*,(n+1)}(x,z) = \text{clamp}_{[0,1]}\left(\frac{(r+\nu)\, U_U^{(m+1)}(x,z) \ - \ \lambda_U\, I^{(m+1)}(x,z)}{P_U(z)\, x}\right),$$

where $I^{(m+1)}(x,z)$ is the tail integral computed in Step 2.3. Anderson acceleration (rank-1) is applied to the vector $\{p^{*,(n+1)}(x,z)\}_x$ before writing it back to cache.

Step 5. Update tightness θ_U by free entry (outer update). Given the updated composition $u_U^{(n+1)}(\cdot,z)$ and the converged frontier surplus $J_U(x,1,z)$, update θ_U using free entry (8):

$$q_U\big(\theta_U^{(n+1)}(z)\big) = \frac{k_U}{\int_0^1 J_U(x,1,z)\, u_U^{(n+1)}(x,z)\ \mathrm{d}x},$$

and invert $q_U(\theta) = \mu_U \theta^{-\eta_U}$ to obtain $\theta_U^{(n+1)}(z)$.
Note: Use Anderson acceleration when updating tightness, to reduce fluctuations.

Step 6. Convergence (outer loop). Stop if

$$\max\Big\{|\theta_U^{(n+1)}(z) - \theta_U^{(n)}(z)|,\ \|p^{*,(n+1)}(\cdot,z) - p^{*,(n)}(\cdot,z)\|_\infty,\ \|u_U^{(n+1)}(\cdot,z) - u_U^{(n)}(\cdot,z)\|_\infty\Big\} < \text{tol}.$$

Otherwise set $n \leftarrow n+1$ and return to Step 2.

Remark (two-block model). In the full model, Step 2 takes $U_S(\cdot,z)$ as given from the skilled block, while the skilled block takes the trained mass $m_S(\cdot,z) = \frac{\phi}{\nu} t(\cdot,z)$ (in steady state) as given from the unskilled block. A global equilibrium is obtained by iterating on the two blocks until these link variables are consistent.

6 Skilled Labor Market (Pissarides Ch. 4: labor turnover and OJS)

This section adapts Chapter 4 ("Labor Turnover and On-the-Job Search") of Pissarides (2000) to our segmented environment with aggregate regime $z \in \mathcal{Z}$ and worker heterogeneity $x \in [0,1]$. The skilled block features (i) idiosyncratic match quality $p \in [0,1]$ that is redrawn by shocks, (ii) endogenous separation via a cutoff $p_S^*(x,z)$, and (iii) on-the-job search (OJS) as a discrete choice with a flow search cost. Workers may quit directly to better jobs (a job ladder) without intervening unemployment.

6.1 Primitives and market objects

Trained population. Only trained workers participate in the skilled market. Let $m_S(x,z)$ be the density of trained workers of type x in regime z. In stationarity,

$$m_S(x,z) = \frac{\phi}{\nu}\, t(x,z),$$

as implied by training completion and demographic turnover.

Production. A skilled match between worker type x and idiosyncratic quality $p \in [0,1]$ produces

$$\pi_S(x,p,z) = P_S(z)\, x\, p.$$

Separations and quality shocks. A filled skilled match is destroyed exogenously at Poisson rate $\xi_S > 0$. In addition, match quality is hit by shocks at Poisson rate $\lambda_S > 0$. Upon a shock, quality is redrawn i.i.d. from Γ_z on $[0,1]$ with density γ_z. A match is endogenously destroyed whenever the post-shock draw is below a reservation cutoff $p_S^*(x,z) \in (0,1)$.

Interpretation of Γ_z. The distribution Γ_z is the *match-quality distribution* in the skilled market. It governs both (i) initial match formation (offers to unemployed workers) and (ii) quality redraws after idiosyncratic shocks. Unlike the unskilled market, skilled matches are not born at a fixed frontier; instead, match quality is drawn from Γ_z at formation and after shocks.

OJS (discrete choice). An employed worker chooses whether to search while employed. Searching incurs a flow cost $\sigma_S \geq 0$ (paid by the worker). If searching, the worker receives outside offers at the same contact rate as an unemployed job seeker, $\theta_S(z)\, q_S(\theta_S(z))$. Each offer is a draw $\tilde{p} \sim \Gamma_z$. The worker quits to the outside job iff the offer is strictly better (job ladder), $\tilde{p} > p$.

Demographic turnover. All workers exit at Poisson rate $\nu > 0$.

6.2 Matching and tightness

Let $u_S(x,z)$ denote the density of skilled unemployed by type, and let $e_S(x,p,z)$ denote the density of employed trained workers by (x,p). Define the *OJS policy* $s^*(x,p,z) \in \{0,1\}$ (defined below), and the aggregate mass of skilled job seekers in regime z as

$$\widetilde{\mathcal{U}}_S(z) \;\equiv\; \underbrace{\int_0^1 u_S(x,z)\,\mathrm{d}x}_{\mathcal{U}_S(z)} \;+\; \underbrace{\int_0^1\!\!\int_0^1 s^*(x,p,z)\, e_S(x,p,z)\,\mathrm{d}p\,\mathrm{d}x}_{\mathcal{E}_S^{\text{search}}(z)}.$$

Let $\mathcal{V}_S(z)$ be the mass of skilled vacancies and define tightness as

$$\theta_S(z) = \frac{\mathcal{V}_S(z)}{\widetilde{\mathcal{U}}_S(z)}.$$

Matching is Cobb–Douglas in vacancies and job seekers:

$$M_S\big(\widetilde{\mathcal{U}}_S(z), \mathcal{V}_S(z)\big) = \mu_S\, \widetilde{\mathcal{U}}_S(z)^{\eta_S} \mathcal{V}_S(z)^{1-\eta_S}, \qquad \eta_S \in (0,1),$$

implying the vacancy contact (filling) rate

$$q_S(\theta_S(z)) = \mu_S\, \theta_S(z)^{-\eta_S},$$

and the contact rate for any job seeker (unemployed or employed searcher)

$$\theta_S(z)\, q_S(\theta_S(z)).$$

6.3 Value functions, bargaining, and two wage schedules

Let

- $U_S(x,z)$ be the value of skilled unemployment;
- $E_S^0(x,p,z)$ and $E_S^1(x,p,z)$ be worker values in a job of quality p without and with OJS;
- $J_S^0(x,p,z)$ and $J_S^1(x,p,z)$ be the corresponding firm values;
- $V_S(z)$ be the value of a skilled vacancy.

Define the worker's continuation value in a job of quality p as

$$E_S(x,p,z) \equiv \max\{E_S^0(x,p,z),\, E_S^1(x,p,z)\}, \qquad J_S(x,p,z) \equiv J_S^{s^*(x,p,z)}(x,p,z).$$

Surplus and Nash bargaining. For each search regime $s \in \{0,1\}$ define match surplus

$$S_S^s(x,p,z) \equiv \Big(E_S^s(x,p,z) - U_S(x,z)\Big) + J_S^s(x,p,z).$$

Nash bargaining with worker weight $\beta_S \in (0,1)$ implies

$$E_S^s(x,p,z) - U_S(x,z) = \beta_S\, S_S^s(x,p,z), \qquad J_S^s(x,p,z) = (1-\beta_S)\, S_S^s(x,p,z), \qquad s \in \{0,1\}. \tag{17}$$

This delivers two wage schedules $w_S^0(x,p,z)$ and $w_S^1(x,p,z)$ because the continuation values differ across s (search changes quit hazards and introduces the search cost).

Unemployed worker. A skilled unemployed worker meets a vacancy at rate $\theta_S(z)q_S(\theta_S(z))$ and draws $\tilde{p} \sim \Gamma_z$. The worker accepts iff the resulting match has nonnegative surplus (equivalently $\tilde{p} \geq p_S^*(x,z)$). New matches start with the worker free to choose whether to search or not, i.e. the acceptance value is $E_S(x,\tilde{p},z)$. The unemployment HJB is

$$(r+\nu)\, U_S(x,z) = b_S(z) + \theta_S(z) q_S(\theta_S(z)) \int_{p_S^*(x,z)}^1 \Big(E_S(x,\tilde{p},z) - U_S(x,z)\Big)\, \mathrm{d}\Gamma_z(\tilde{p}). \tag{18}$$

Employed worker, no search ($s = 0$). With no search, the only events are demographic exit ν, exogenous separation ξ_S, and quality shocks λ_S. The worker HJB is

$$(r + \nu + \xi_S + \lambda_S)\, E_S^0(x, p, z) = w_S^0(x, p, z) + \xi_S\, U_S(x, z) + \lambda_S \left[U_S(x, z)\, \Gamma_z\big(p_S^*(x, z)\big) + \int_{p_S^*(x,z)}^1 E_S(x, p', z)\, \mathrm{d}\Gamma_z(p') \right]. \tag{19}$$

Employed worker, search ($s = 1$). Searching costs σ_S per unit time and generates outside contacts at rate $\theta_S(z) q_S(\theta_S(z))$. Upon an offer $\tilde{p}$, the worker quits iff $\tilde{p} > p$ and the offered match is feasible. Hence

$$\begin{aligned}(r + \nu + \xi_S + \lambda_S)\, E_S^1(x, p, z) = {} & \big(w_S^1(x, p, z) - \sigma_S\big) + \xi_S\, U_S(x, z) \\ & + \lambda_S \left[U_S(x, z)\, \Gamma_z\big(p_S^*(x, z)\big) + \int_{p_S^*(x,z)}^1 E_S(x, p', z)\, \mathrm{d}\Gamma_z(p') \right] \\ & + \theta_S(z) q_S(\theta_S(z)) \int_{\max\{p, p_S^*(x,z)\}}^1 \Big(E_S(x, \tilde{p}, z) - E_S^1(x, p, z) \Big)\, \mathrm{d}\Gamma_z(\tilde{p}).\end{aligned} \tag{20}$$

The last term is the continuous-time job ladder: only improvements ($\tilde{p} > p$) induce quits.

Firm values: quits matter only under OJS. A filled job yields flow profit $P_S(z)\, x\, p - w_S^s(x, p, z)$. With no search there is no quit hazard. With OJS, the job is lost at rate $\theta_S(z) q_S(\theta_S(z)) \big(1 - \Gamma_z(p)\big)$ because any offer above p triggers a quit. Thus:

$$(r + \nu + \xi_S + \lambda_S)\, J_S^0(x, p, z) = P_S(z)\, x\, p - w_S^0(x, p, z) + \lambda_S \int_{p_S^*(x,z)}^1 J_S(x, p', z)\, \mathrm{d}\Gamma_z(p'), \tag{21}$$

$$\begin{aligned}\Big(r + \nu + \xi_S + \lambda_S + \theta_S(z) q_S(\theta_S(z)) \big(1 - \Gamma_z(p)\big) \Big)\, J_S^1(x, p, z) = {} & P_S(z)\, x\, p - w_S^1(x, p, z) \\ & + \lambda_S \int_{p_S^*(x,z)}^1 J_S(x, p', z)\, \mathrm{d}\Gamma_z(p').\end{aligned} \tag{22}$$

6.4 Policy rules: endogenous separation and OJS cutoff

Endogenous separation cutoff. Define overall surplus as

$$S_S(x, p, z) \equiv \max\{S_S^0(x, p, z),\, S_S^1(x, p, z)\}.$$

Endogenous separation is characterized by a cutoff $p_S^*(x, z)$ satisfying

$$S_S\big(x, p_S^*(x, z), z\big) = 0, \qquad S_S(x, p, z) \geq 0 \iff p \geq p_S^*(x, z). \tag{23}$$

OJS cutoff (search vs no-search). The worker searches iff it is optimal:

$$s^*(x, p, z) = \mathbb{1}\{E_S^1(x, p, z) \geq E_S^0(x, p, z)\}.$$

Under monotonicity in p, there exists an OJS cutoff $p_S^{\mathrm{oj}}(x, z) \geq p_S^*(x, z)$ such that

$$E_S^1(x, p, z) \geq E_S^0(x, p, z) \iff p < p_S^{\mathrm{oj}}(x, z), \qquad E_S^1\big(x, p_S^{\mathrm{oj}}(x, z), z\big) = E_S^0\big(x, p_S^{\mathrm{oj}}(x, z), z\big). \tag{24}$$

6.5 Vacancies, poaching, and free entry (Chapter 4 form)

Posting a skilled vacancy costs $k_S > 0$ per unit time. A vacancy meets a *random job seeker* at rate $q_S(\theta_S(z))$. Since job seekers include unemployed and employed searchers, the expected value of a contact depends on the composition of seekers.

Define the density of job seekers (unemployed + employed searchers) as

$$\widetilde{u}_S(x,z) \equiv u_S(x,z) + \int_0^1 s^*(x,p,z)\, e_S(x,p,z)\ \mathrm{d}p, \qquad \widetilde{\mathcal{U}}_S(z) = \int_0^1 \widetilde{u}_S(x,z)\ \mathrm{d}x.$$

Conditional on a contact, the worker type is drawn from $\widetilde{u}_S(\cdot, z)$, and:

- if the contacted seeker is unemployed, the match quality draw is $\tilde{p} \sim \Gamma_z$ and the match is formed iff $\tilde{p} \geq p_S^*(x,z)$, yielding value $J_S(x,\tilde{p},z)$;
- if the contacted seeker is an employed searcher currently at quality p, the offer is $\tilde{p} \sim \Gamma_z$ and the worker accepts (quits) iff $\tilde{p} > p$ and $\tilde{p} \geq p_S^*(x,z)$, yielding value $J_S(x,\tilde{p},z)$.

Thus the vacancy HJB is

$$rV_S(z) = -k_S + q_S(\theta_S(z))\Big(\mathcal{J}_S(z) - V_S(z)\Big), \tag{25}$$

where the expected surplus from a random contact is

$$\begin{aligned}\mathcal{J}_S(z) \equiv \frac{1}{\widetilde{\mathcal{U}}_S(z)}\Bigg[&\underbrace{\int_0^1 u_S(x,z) \int_{p_S^*(x,z)}^1 J_S(x,\tilde{p},z)\ \mathrm{d}\Gamma_z(\tilde{p})\ \mathrm{d}x}_{\text{meeting unemployed}} \\ &+ \underbrace{\int_0^1 \int_0^1 s^*(x,p,z)\, e_S(x,p,z) \int_{\max\{p,p_S^*(x,z)\}}^1 J_S(x,\tilde{p},z)\ \mathrm{d}\Gamma_z(\tilde{p})\ \mathrm{d}p\ \mathrm{d}x}_{\text{poaching employed searchers}}\Bigg].\end{aligned} \tag{26}$$

Free entry implies $V_S(z) = 0$, hence the Chapter 4 free-entry restriction

$$q_S(\theta_S(z))\, \mathcal{J}_S(z) = k_S. \tag{27}$$

Given (u_S, e_S, s^*), this pins down $\theta_S(z)$ by inverting q_S.

6.5.1 Closed-form surplus equations for $s = 0$ and $s = 1$

Throughout, define the skilled offer CDF $\Gamma_z(\cdot)$ and write

$$\kappa_S(z) \equiv \theta_S(z)\, q_S(\theta_S(z)).$$

Recall the regime-s surplus and Nash split:

$$S_S^s(x,p,z) \equiv \big(E_S^s(x,p,z) - U_S(x,z)\big) + J_S^s(x,p,z), \qquad E_S^s - U_S = \beta_S S_S^s, \qquad J_S^s = (1-\beta_S) S_S^s,$$

and the *overall* (optimal-regime) surplus

$$S_S(x,p,z) \equiv \max\{S_S^0(x,p,z), S_S^1(x,p,z)\}.$$

Step 1: a useful identity for continuation terms. Because $E_S(x,p,z) = U_S(x,z) + \beta_S S_S(x,p,z)$ and $J_S(x,p,z) = (1-\beta_S)S_S(x,p,z)$ under the optimal regime at (x,p,z), we have

$$E_S(x,p,z) + J_S(x,p,z) = U_S(x,z) + S_S(x,p,z).$$

Hence, for any lower bound $\underline{p} \geq p_S^*(x,z)$,

$$\int_{\underline{p}}^1 \big(E_S(x,p',z) + J_S(x,p',z)\big)\, d\Gamma_z(p') = \big(1-\Gamma_z(\underline{p})\big)\, U_S(x,z) + \int_{\underline{p}}^1 S_S(x,p',z)\, d\Gamma_z(p').$$

Step 2: eliminate wages by adding worker and firm HJBs. Take the worker HJB in regime s and subtract the unemployment HJB to obtain an equation for $(E_S^s - U_S)$. Add the firm HJB for J_S^s. The wage w_S^s cancels identically, leaving an HJB for S_S^s.

Surplus with no search ($s=0$). Adding (19) (after subtracting U_S) and (21) gives

$$(r+\nu+\xi_S+\lambda_S)\, S_S^0(x,p,z) = P_S(z)\, x\, p - (r+\nu)\, U_S(x,z) + \lambda_S \int_{p_S^*(x,z)}^1 S_S(x,p',z)\, d\Gamma_z(p').$$

Now use the unemployment HJB (18) and Nash:

$$\begin{aligned}(r+\nu)\, U_S(x,z) &= b_S(z) + \kappa_S(z) \int_{p_S^*(x,z)}^1 \big(E_S(x,p',z) - U_S(x,z)\big)\, d\Gamma_z(p') \\ &= b_S(z) + \beta_S\, \kappa_S(z) \int_{p_S^*(x,z)}^1 S_S(x,p',z)\, d\Gamma_z(p').\end{aligned}$$

Substituting yields the closed form

$$(r+\nu+\xi_S+\lambda_S)\, S_S^0(x,p,z) = P_S(z)\, x\, p - b_S(z) + \Big(\lambda_S - \beta_S\, \kappa_S(z)\Big) \int_{p_S^*(x,z)}^1 S_S(x,p',z)\, d\Gamma_z(p'). \tag{28}$$

Surplus with on-the-job search ($s=1$). Proceeding analogously with (20) and (22) gives (before simplification)

$$\begin{aligned}&\big(r+\nu+\xi_S+\lambda_S\big)\, S_S^1(x,p,z) \;+\; \kappa_S(z)\big(1-\Gamma_z(p)\big)\, J_S^1(x,p,z) \\ &\qquad = P_S(z)\, x\, p - (r+\nu)\, U_S(x,z) - \sigma_S + \lambda_S \int_{p_S^*(x,z)}^1 S_S(x,p',z)\, d\Gamma_z(p') \\ &\qquad\quad + \kappa_S(z) \int_{\max\{p,p_S^*(x,z)\}}^1 \big(E_S(x,\tilde{p},z) - E_S^1(x,p,z)\big)\, d\Gamma_z(\tilde{p}).\end{aligned}$$

Use Nash $J_S^1 = (1-\beta_S)S_S^1$ and

$$E_S(x,\tilde{p},z) - E_S^1(x,p,z) = \big(U_S + \beta_S S_S(x,\tilde{p},z)\big) - \big(U_S + \beta_S S_S^1(x,p,z)\big) = \beta_S\Big(S_S(x,\tilde{p},z) - S_S^1(x,p,z)\Big),$$

together with (18) as above. For feasible jobs $p \geq p_S^*(x,z)$, $\max\{p,p_S^*\} = p$, and the $-S_S^1$ part of the ladder term combines with the firm quit term to produce a *full* additional discount rate $\kappa_S(z)\big(1-\Gamma_z(p)\big)$ on S_S^1. The result is the closed form

$$\begin{aligned}\Big(r+\nu+\xi_S+\lambda_S+\kappa_S(z)\big(1-\Gamma_z(p)\big)\Big)\, S_S^1(x,p,z) &= P_S(z)\, x\, p - b_S(z) - \sigma_S \\ +\lambda_S \int_{p_S^*(x,z)}^1 S_S(x,p',z)\, d\Gamma_z(p') &- \beta_S\, \kappa_S(z) \int_{p_S^*(x,z)}^p S_S(x,p',z)\, d\Gamma_z(p').\end{aligned} \tag{29}$$

Interpretation. Equation (28) is the standard turnover/quality-shock surplus equation: flow output minus unemployment flow value, plus expected continuation surplus under quality shocks, with the worker capturing a fraction β_S of the unemployment-contact option via the term $\beta_S \kappa_S \int S_S \, d\Gamma_z$. Equation (29) adds (i) the search cost $-\sigma_S$, (ii) an additional effective discount from quits at rate $\kappa_S(z)\big(1 - \Gamma_z(p)\big)$, and (iii) a ladder term $-\beta_S \kappa_S \int_{p^*}^{p} S_S \, d\Gamma_z$ reflecting that under OJS the worker can appropriate part of the match rents through outside offers below the current p threshold.

Cutoffs. Given $S_S(x,p,z) = \max\{S_S^0, S_S^1\}$, the endogenous separation cutoff satisfies $S_S(x, p_S^*(x,z), z) = 0$. The OJS cutoff $p_S^{\mathrm{oj}}(x,z)$ is pinned down by $S_S^1(x, p_S^{\mathrm{oj}}, z) = S_S^0(x, p_S^{\mathrm{oj}}, z)$ (equivalently $E_S^1 = E_S^0$).

Closed-form wage schedules (skilled). Wages are recovered by combining each firm HJB with the corresponding closed-form surplus equation, using Nash $J_S^s = (1-\beta_S) S_S^s$. The wage terms cancel between the worker and firm HJBs, and the surplus equations (28)–(29) supply the remainder.

No OJS ($s = 0$). Multiplying (28) by $(1-\beta_S)$ and comparing with the firm HJB (21) yields

$$w_S^0(x,p,z) = \beta_S \, P_S(z) \, x \, p \; + \; (1-\beta_S) \, b_S(z) \; + \; \beta_S (1-\beta_S) \, \kappa_S(z) \underbrace{\int_{p_S^*(x,z)}^{1} S_S(x,p',z) \; \mathrm{d}\Gamma_z(p')}_{=:\, I(x,z)} . \tag{30}$$

The wage is linear in p for fixed x: the third term is an x-specific constant that reflects the unemployment-contact option available to the worker (weight β_S) and surrendered to the firm (weight $1-\beta_S$) when the match is valued under no search.

OJS ($s = 1$). The same procedure applied to (29) and (22) gives

$$w_S^1(x,p,z) = \beta_S \, P_S(z) \, x \, p + (1-\beta_S)\big(b_S(z) + \sigma_S\big) + \beta_S (1-\beta_S) \, \kappa_S(z) \underbrace{\int_{p_S^*(x,z)}^{p} S_S(x,p',z) \; \mathrm{d}\Gamma_z(p')}_{=:\, I_{\mathrm{low}}(x,p,z)} . \tag{31}$$

Comparing (30) and (31), the OJS wage premium is

$$w_S^1(x,p,z) - w_S^0(x,p,z) = -(1-\beta_S) \, \sigma_S \; - \; \beta_S (1-\beta_S) \, \kappa_S(z) \int_{p}^{1} S_S(x,p',z) \; \mathrm{d}\Gamma_z(p'). \tag{32}$$

The premium is *non-positive*: the search cost $(1-\beta_S)\sigma_S$ is borne entirely by the worker (Nash shifts it from w_S^1 relative to w_S^0), while the ladder option term $-\beta_S(1-\beta_S)\kappa_S \int_p^1 S_S \; \mathrm{d}\Gamma_z \le 0$ reflects the firm's increased quit risk under OJS — the firm is compensated by a wage discount proportional to the expected capital gain it would lose to a poaching offer. Both terms vanish as $p \to 1$ (no further upward ladder) and are largest near $p_S^*(x,z)$.

6.6 Stationary cross-sectional distributions (trained workers)

Let $u_S(x,z)$ be the density of skilled unemployed and $e_S(x,p,z)$ the density of employed trained workers at quality p. Accounting within the trained population:

$$m_S(x,z) = u_S(x,z) + \int_{p_S^*(x,z)}^{1} e_S(x,p,z) \; \mathrm{d}p.$$

In the Chapter 4 (OJS) environment, employed searchers are *job seekers* and generate *job-to-job* moves. A quit is therefore a separation *from the current firm* but *not* a transition into unemployment. For this reason, quits enter (i) the employed flow balance across p and (ii) vacancy values, but they do not appear as inflows into u_S.

For computation we discretize p and solve the implied linear flow-balance system. In density form, the key hazards are:

Endogenous destruction from quality shocks. Quality shocks arrive at rate λ_S and redraw $p' \sim \Gamma_z$. A match is destroyed if $p' < p_S^*(x,z)$, so the endogenous destruction hazard is

$$\delta_S^{\text{end}}(x,z) \equiv \lambda_S\, \Gamma_z\big(p_S^*(x,z)\big).$$

Job-to-job quit hazard under OJS. If the worker searches ($s^*(x,p,z) = 1$), outside offers arrive at rate $\theta_S(z)q_S(\theta_S(z))$ and the worker quits iff the offer draw $\tilde{p}$ exceeds the current p, which occurs with probability $1 - \Gamma_z(p)$. Hence the job-to-job quit hazard is

$$\delta_S^{\text{jj}}(x,p,z) \equiv s^*(x,p,z)\, \theta_S(z)q_S(\theta_S(z))\big(1 - \Gamma_z(p)\big).$$

This is a *within-employment* reallocation hazard: it removes mass from low-p jobs and reallocates it to higher-p jobs.

Unemployment balance. Unemployment receives inflow from training completion and from separations that create unemployment (exogenous separations and endogenous destruction). Unemployment loses mass through demographic exit and job finding into acceptable matches ($\tilde{p} \geq p_S^*$). Thus, for each x,

$$\Big(\nu + \theta_S(z)q_S(\theta_S(z))\,(1 - \Gamma_z(p_S^*(x,z)))\Big)\, u_S(x,z) = \phi\, t(x,z) + \int_{p_S^*(x,z)}^{1} \big(\xi_S + \delta_S^{\text{end}}(x,z)\big)\, e_S(x,p,z)\ \mathrm{d}p. \tag{33}$$

Employment-by-quality balance. For $p \geq p_S^*(x,z)$, the density $e_S(x,p,z)$ satisfies a steady-state balance: outflows from the state (x,p) equal inflows into (x,p). Outflows are due to demographic exit, exogenous separation, quality shocks (which move the match to a new p'), and job-to-job quits. Inflows come from three sources: (i) hiring from unemployment into quality p (offer density $\gamma_z(p)$), (ii) quality shocks from any employed state into p (redraw density $\gamma_z(p)$), and (iii) job-to-job upgrades from worse jobs $p' < p$ when those workers search and draw p.

Define the job-to-job quit hazard δ_S^{jj} as above. Then, for each x and each $p \geq p_S^*(x,z)$,

$$\begin{aligned}\Big(\nu + \xi_S + \lambda_S + \delta_S^{\text{jj}}(x,p,z)\Big)\, e_S(x,p,z) = {} & \underbrace{\theta_S(z)q_S(\theta_S(z))\, u_S(x,z)\, \gamma_z(p)}_{\text{hiring from unemployment into } p} \\ & + \underbrace{\lambda_S\, \gamma_z(p) \int_{p_S^*(x,z)}^{1} e_S(x,p',z)\ \mathrm{d}p'}_{\text{quality shocks into } p} \\ & + \underbrace{\theta_S(z)q_S(\theta_S(z))\, \gamma_z(p) \int_{p_S^*(x,z)}^{p} s^*(x,p',z)\, e_S(x,p',z)\ \mathrm{d}p'}_{\text{job-to-job inflow from worse jobs}}.\end{aligned} \tag{34}$$

The last term is the job ladder: mass leaves worse jobs via job-to-job quits and arrives in better jobs according to the offer density γ_z.

Implementation remark. On a grid $p_1 < \cdots < p_{N_p}$, (34) becomes a linear system in the unknown vector $\{e_S(x, p_i, z)\}_{i=1}^{N_p}$ for each x: the quit hazard contributes to the diagonal; quality shocks add a rank-one inflow term proportional to $\gamma_z(p_i)$; and the job-ladder inflow term is lower-triangular because it depends only on $\{e_S(x, p_j, z)\}_{j<i}$ in the OJS region ($s^* = 1$).

6.7 Skilled-block equilibrium and nested fixed-point algorithm

Fix a regime $z \in \mathcal{Z}$ and take the trained population density $m_S(\cdot, z)$ as given from the unskilled block. A *skilled-block equilibrium* is a collection

$$\Big\{\theta_S(z),\ U_S(\cdot, z),\ E_S^0(\cdot,\cdot,z), E_S^1(\cdot,\cdot,z),\ J_S^0(\cdot,\cdot,z), J_S^1(\cdot,\cdot,z),\ w_S^0(\cdot,\cdot,z), w_S^1(\cdot,\cdot,z),$$

$$p_S^*(\cdot, z),\ p_S^{\mathrm{oj}}(\cdot, z),\ s^*(\cdot,\cdot,z),\ u_S(\cdot, z),\ e_S(\cdot,\cdot,z)\Big\}$$

such that the following conditions hold:

(S1) **Worker values.** Unemployment satisfies (18). Employed values satisfy (19)–(20), with

$$E_S(x, p, z) \equiv \max\{E_S^0(x, p, z),\ E_S^1(x, p, z)\}.$$

(S2) **Firm values.** Filled-job values satisfy (21)–(22), with

$$J_S(x, p, z) \equiv J_S^{s^*(x,p,z)}(x, p, z).$$

(S3) **OJS policy.** The worker's search decision is optimal:

$$s^*(x, p, z) = \mathbb{1}\{E_S^1(x, p, z) \geq E_S^0(x, p, z)\},$$

and under monotonicity it is summarized by a cutoff $p_S^{\mathrm{oj}}(x, z)$ satisfying (24).

(S4) **Endogenous separation.** The reservation cutoff $p_S^*(x, z)$ satisfies (23).

(S5) **Wage determination.** Wages satisfy Nash bargaining in the sense that they deliver the intended surplus split, implemented via (17) together with the HJBs (19)–(20). (Equivalently, w_S^{s} are recovered from the worker HJBs and then checked/adjusted to enforce (17) pointwise.)

(S6) **Tightness and free entry (Chapter 4 congestion).** Tightness is defined using job seekers (unemployed plus employed searchers) and free entry holds:

$$\theta_S(z) = \frac{\mathcal{V}_S(z)}{\widetilde{\mathcal{U}}_S(z)}, \qquad q_S(\theta_S(z))\, \mathcal{J}_S(z) = k_S,$$

with $\widetilde{\mathcal{U}}_S(z)$ defined in the matching subsection and $\mathcal{J}_S(z)$ in (26).

(S7) **Stationary distributions.** Given $(\theta_S, p_S^*, p_S^{\mathrm{oj}}, s^*)$, the stationary densities (u_S, e_S) satisfy (33)–(34) and the accounting identity $m_S(x, z) = u_S(x, z) + \int_{p_S^*(x,z)}^1 e_S(x, p, z)\, \mathrm{d}p$.

Nested fixed-point algorithm (skilled side; fixed regime z). For computation it is convenient to organize the skilled block as a nested fixed point. The *inner loop* solves the value functions (and wage schedules) given outer objects. The *outer loop* updates $(\theta_S, p_S^*, p_S^{\mathrm{oj}})$ and the stationary composition until all equilibrium conditions hold jointly.

Outer-loop unknowns. Iterate on

$$\Big(\theta_S^{(n)}(z),\ p_S^{*,(n)}(\cdot,z),\ p_S^{\mathrm{oj},(n)}(\cdot,z)\Big).$$

Step 1. Initialize (outer loop). Choose initial guesses $\theta_S^{(0)}(z) > 0$ and feasible cutoffs $p_S^{*,(0)}(x,z) \in (0,1)$ and $p_S^{\mathrm{oj},(0)}(x,z) \in [p_S^{*,(0)}(x,z), 1]$. Initialize stationary densities $(u_S^{(0)}, e_S^{(0)})$ consistent with $m_S(\cdot,z)$ (e.g. set $u_S^{(0)} = \kappa\, m_S$ and allocate the remainder of mass to the top p grid point).

Step 2. Inner loop: surplus iteration (given outer objects). Holding $\big(\theta_S^{(n)},\, p_S^{*,(n)},\, p_S^{\mathrm{oj},(n)}\big)$ fixed, iterate on $(U_S,\, S_S^0,\, S_S^1)$ until convergence. Let $\kappa_S = \theta_S^{(n)} q_S(\theta_S^{(n)})$ throughout. One inner-loop cycle for each x:

Step 2.1. Tail integral. Let j_0 be the first grid index with $p_{j_0} \geq p_S^{*,(n)}(x,z)$. Compute the tail integral using the *previous* iteration's surplus:

$$I^{(m)}(x,z) = \int_{p_S^{*,(n)}(x,z)}^{1} S_S^{\max,(m)}(x,p',z)\ \mathrm{d}\Gamma_z(p'), \qquad S_S^{\max,(m)} = \max\big\{S_S^{0,(m)},\, S_S^{1,(m)}\big\}.$$

This is approximated by accumulating the weighted sum $\sum_{k\geq j_0} S_S^{\max,(m)}(x,p_k,z)\, w_\Gamma(p_k)$ in a backward pass over the p-grid.

Step 2.2. Unemployment value (closed form).

$$U_S^{(m+1)}(x,z) = \frac{b_S(z) + \kappa_S\, \beta_S\, I^{(m)}(x,z)}{r+\nu}.$$

Step 2.3. Surplus on p-grid (closed-form evaluation). For each grid point p_j:

- If $j < j_0$ (infeasible region): set $S_S^{0,(m+1)} = S_S^{1,(m+1)} = 0$.
- If $j \geq j_0$:

$$S_S^{0,(m+1)}(x,p_j,z) = \frac{P_S(z)\, x\, p_j - (r+\nu)\, U_S^{(m+1)} + \lambda_S\, I^{(m)}}{r+\nu+\xi_S+\lambda_S},$$

$$S_S^{1,(m+1)}(x,p_j,z) = \frac{P_S(z)\, x\, p_j - (r+\nu)\, U_S^{(m+1)} - \sigma_S + \lambda_S\, I^{(m)} + \kappa_S\, \beta_S \displaystyle\int_{p_j}^{1} S_S^{\max,(m)}\ \mathrm{d}\Gamma_z}{r+\nu+\xi_S+\lambda_S+\kappa_S\big(1-\Gamma_z(p_j)\big)}.$$

The numerator integral $\int_{p_j}^{1} S_S^{\max,(m)}\ \mathrm{d}\Gamma_z = \texttt{tailE}[j]$ is available from Step 2.1.

Step 2.4. Nash split. Recover employed values and firm values via $E_S^{s,(m+1)} = U_S^{(m+1)} + \beta_S S_S^{s,(m+1)}$, $\quad J_S^{s,(m+1)} = (1-\beta_S) S_S^{s,(m+1)}$, $\quad s \in \{0,1\}$. Wages are not computed explicitly.

Stop when $\sup_{x,p} \max\big\{|S_S^{0,(m+1)} - S_S^{0,(m)}|,\, |S_S^{1,(m+1)} - S_S^{1,(m)}|,\, |U_S^{(m+1)} - U_S^{(m)}|\big\} < \texttt{tol_inner}$.

Step 3. Update cutoffs (outer update).

Step 3.1. Separation cutoff p_S^*. Scan forward from the *previous* cutoff index j_0^{prev} for the first grid point where $S_S^{\max,(m+1)}(x, p_j, z) \geq 0$:

$$p_S^{*,\text{prop}}(x, z) = p_{j^*}, \qquad j^* = \min\{j \geq j_0^{\text{prev}} : S_S^{\max}(x, p_j, z) \geq 0\}.$$

Apply damped update with damping coefficient $\omega \in (0, 1)$:

$$p_S^{*,(n+1)}(x, z) = \omega\, p_S^{*,\text{prop}}(x, z) + (1 - \omega)\, p_S^{*,(n)}(x, z).$$

Step 3.2. OJS cutoff p_S^{oj}. Scan forward from j_0 for the first grid point where $S_S^{1,(m+1)}(x, p_j, z) - S_S^{0,(m+1)}(x, p_j, z) \leq 0$:

$$p_S^{\text{oj},(n+1)}(x, z) = p_{j^{\text{oj}}}, \qquad j^{\text{oj}} = \min\{j \geq j_0 : S_S^1(x, p_j, z) \leq S_S^0(x, p_j, z)\}.$$

Apply the same damped update. The OJS policy follows as $s^*(x, p, z) = \mathbf{1}\{p < p_S^{\text{oj},(n+1)}(x, z)\}$.

Step 4. Update stationary distributions (composition block).
Given $\big(\theta_S^{(n)}, p_S^{*,(n+1)}, p_S^{\text{oj},(n+1)}\big)$, solve the stationary flow-balance system (33)–(34) (on the discretized p grid) subject to the accounting identity for each x. From the solution, compute the seeker density

$$\widetilde{u}_S(x, z) = u_S(x, z) + \int_0^1 s^*(x, p, z)\, e_S(x, p, z)\, \mathrm{d}p, \qquad \widetilde{\mathcal{U}}_S(z) = \int_0^1 \widetilde{u}_S(x, z)\, \mathrm{d}x.$$

Step 5. Update tightness by free entry (outer update). Compute $\mathcal{J}_S(z)$ from (26) using the updated (u_S, e_S, s^*) and the converged $J_S(x, p, z)$. Update θ_S using free entry (27):

$$q_S(\theta_S^{(n+1)}(z)) = \frac{k_S}{\mathcal{J}_S(z)}.$$

Invert $q_S(\theta) = \mu_S \theta^{-\eta_S}$ to obtain $\theta_S^{(n+1)}(z)$.
Note: Use Anderson acceleration when updating tightness, to reduce fluctuations.

Step 6. Convergence (outer loop). Stop if

$$\max\Big\{|\theta_S^{(n+1)}(z) - \theta_S^{(n)}(z)|,\ \|p_S^{*,(n+1)}(\cdot, z) - p_S^{*,(n)}(\cdot, z)\|_\infty,\ \|p_S^{\text{oj},(n+1)}(\cdot, z) - p_S^{\text{oj},(n)}(\cdot, z)\|_\infty\Big\} < \text{tol}.$$

Otherwise set $n \leftarrow n + 1$ and return to Step 2.

Link to the unskilled block. On convergence, the skilled unemployment value $U_S(x, z)$ feeds back into the training value equation (1), while the trained density $m_S(\cdot, z)$ is taken as given from training completion in the unskilled block. A global equilibrium is obtained by iterating between the two blocks until these link variables are jointly consistent.

7 Full Model Equilibrium and Global Algorithm

Fix a regime $z \in \mathcal{Z}$. A *full equilibrium* in regime z consists of unskilled-block and skilled-block objects such that:

(F1) The unskilled block satisfies conditions (E1)–(E5) given the skilled unemployment value $U_S(\cdot, z)$.

(F2) The skilled block satisfies conditions (S1)–(S7) given the trained population density

$$m_S(x, z) = \frac{\phi}{\nu} t(x, z).$$

(F3) The link variables are consistent:

$$U_S(\cdot, z) \text{ used in the unskilled block equals the skilled-block solution,}$$

$$m_S(\cdot, z) \text{ used in the skilled block equals training completion from the unskilled block.}$$

Global fixed-point algorithm (fixed regime z). The full model is solved via an outer fixed point over the link variables $(U_S(\cdot, z), m_S(\cdot, z))$:

Step 1. Initialize. Choose initial guesses $U_S^{(0)}(\cdot, z)$ and $m_S^{(0)}(\cdot, z)$.

Step 2. Solve the unskilled block. Given $U_S^{(n)}(\cdot, z)$, solve the unskilled equilibrium as in Section 5.8. Obtain $(t^{(n)}(\cdot, z), u_U^{(n)}(\cdot, z), \theta_U^{(n)}(z), \dots)$.

Update the implied trained mass:

$$m_S^{\text{new}}(x, z) = \frac{\phi}{\nu} t^{(n)}(x, z).$$

Step 3. Solve the skilled block. Given $m_S^{\text{new}}(\cdot, z)$, solve the skilled equilibrium using the nested algorithm described above. Obtain $U_S^{\text{new}}(\cdot, z)$ and $\theta_S^{(n)}(z)$.

Step 4. Update link variables. Update U_S and m_S.

Step 5. Convergence. Stop if

$$\max\left\{\|U_S^{(n+1)} - U_S^{(n)}\|_\infty,\ \|m_S^{(n+1)} - m_S^{(n)}\|_\infty\right\} < \text{tol}.$$

Otherwise set $n \leftarrow n + 1$ and return to Step 2.

Regime transitions. For each regime $z \in \mathcal{Z}$, compute the stationary equilibrium as above. After an unexpected regime switch $z_0 \to z_1$, value functions and policies jump to the equilibrium associated with z_1, while the cross-sectional densities evolve according to the transition laws in the unskilled and skilled blocks, using the new regime's policy and hazard objects.

8 Model dynamics: evolution of densities after a regime change

Consider an unexpected, permanent regime switch at calendar time T from z_0 to z_1. At $t = T^+$, cross-sectional densities are predetermined by the pre-switch history. For $t > T$, all hazards and policies are those of regime z_1.

Throughout this section, fix $z = z_1$ and suppress z in notation when convenient.

8.1 Time-dependent objects and hazards

Let

- $u_U(x,t)$ (untrained unemployed in unskilled market),
- $t(x,t)$ (in training),
- $u_S(x,t)$ (trained unemployed in skilled market).

Let the unskilled job-finding rate be

$$f_U \equiv \theta_U(z)\, q_U(\theta_U(z)),$$

and the skilled job-finding rate be

$$f_S \equiv \theta_S(z)\, q_S(\theta_S(z)).$$

Training entry is governed by the post-switch policy

$$\tau(x) \in \{0,1\},$$

training completes at rate ϕ, and demographic exit occurs at rate ν.

Unskilled endogenous separations. In the unskilled market, employed untrained matches receive damage shocks at rate λ_U, with post-shock $p' \sim G_z$. The reservation rule $p^*(x)$ implies the endogenous job-destruction hazard

$$\delta_U(x) \equiv \lambda_U\, G_z\big(p^*(x)\big).$$

Skilled endogenous destruction. In the skilled market, quality shocks arrive at rate λ_S and redraw $p' \sim \Gamma_z$. The reservation rule $p_S^*(x)$ implies the endogenous destruction hazard

$$\delta_S^{\text{end}}(x) \equiv \lambda_S\, \Gamma_z\big(p_S^*(x)\big).$$

Exogenous separations occur at rate ξ_S.

8.2 Segment accounting identities

Define the untrained (unskilled-segment) density by type

$$m_U(x,t) \equiv u_U(x,t) + t(x,t) + e_U(x,t),$$

where $e_U(x,t)$ denotes untrained unskilled employment density (not tracked as a separate state variable). Thus,

$$e_U(x,t) = m_U(x,t) - u_U(x,t) - t(x,t).$$

Define the trained (skilled-segment) density by type

$$m_S(x,t) \equiv u_S(x,t) + e_S^{\text{tot}}(x,t),$$

where

$$e_S^{\text{tot}}(x,t) \equiv \int_{p_S^*(x)}^1 e_S(x,p,t)\ \mathrm{d}p$$

is total skilled employment mass of type x (integrating over match quality). We do not need the full p-density to write the laws of motion below, but e_S^{tot} enters skilled-unemployment inflows via separations.

Aggregate segment masses are

$$M_U(t) \equiv \int_0^1 m_U(x,t)\ \mathrm{d}x, \qquad M_S(t) \equiv \int_0^1 m_S(x,t)\ \mathrm{d}x.$$

8.3 Dynamics in the unskilled segment

Training stock. Training receives inflow from untrained unemployment at rate $\tau(x)$ and loses mass through completion at rate ϕ and demographic exit at rate ν:

$$\partial_t t(x,t) = \tau(x)\, u_U(x,t) - (\phi + \nu)\, t(x,t). \tag{35}$$

Untrained unemployment. Untrained unemployment receives inflow from demographic replacement and from endogenous job destruction out of unskilled employment. It loses mass due to job finding, training entry, and demographic exit:

$$\partial_t u_U(x,t) = \nu\, \ell(x) + \delta_U(x)\, e_U(x,t) - \big(f_U + \tau(x) + \nu\big)\, u_U(x,t), \tag{36}$$

where $e_U(x,t) = m_U(x,t) - u_U(x,t) - t(x,t)$.

Untrained-segment mass by type. Because training completion moves workers from the unskilled to the skilled segment, the untrained mass $m_U(x,t)$ evolves as

$$\partial_t m_U(x,t) = \nu\, \ell(x) - \nu\, m_U(x,t) - \phi\, t(x,t). \tag{37}$$

Training entry reshuffles mass within the unskilled segment, so it does not appear in $\partial_t m_U$; completion removes mass from m_U. Note that the outflow from training in (37) is $-\phi\, t$, not $-(\phi+\nu)\, t$: the $-\nu\, m_U(x,t)$ term already accounts for demographic exit from *all* unskilled states, including trainees. The term $-(\phi+\nu)\, t$ in (35) correctly reflects that the training stock loses mass both via completion and via demographic exit; the two equations are consistent because demographic exit of a trainee is captured by $-\nu m_U$ at the segment level and by $-\nu t$ inside $-(\phi+\nu)t$ at the stock level.

8.4 Dynamics in the skilled segment

Skilled unemployment. Skilled unemployment receives inflow from training completion and from separations out of skilled employment that create unemployment (exogenous separations and endogenous destruction). It loses mass due to job finding into acceptable matches and demographic exit:

$$\partial_t u_S(x,t) = \phi\, t(x,t) + \big(\xi_S + \delta_S^{\text{end}}(x)\big)\, e_S^{\text{tot}}(x,t) - \Big(\nu + f_S\big(1 - \Gamma_z(p_S^*(x))\big)\Big)\, u_S(x,t). \tag{38}$$

Trained-segment mass by type. The trained mass $m_S(x,t)$ increases via training completion and decreases via demographic exit:

$$\partial_t m_S(x,t) = \phi\, t(x,t) - \nu\, m_S(x,t). \tag{39}$$

(Job-to-job moves reshuffle employment across firms and match qualities but do not change segment membership, hence do not enter $\partial_t m_S$.)

8.5 Aggregate masses: unskilled and skilled segments

Integrating (37)–(39) over $x \in [0,1]$ yields the aggregate segment mass dynamics:

$$\dot{M}_U(t) = \nu \int_0^1 \ell(x)\ \mathrm{d}x - \nu M_U(t) - \phi \int_0^1 t(x,t)\ \mathrm{d}x, \tag{40}$$

$$\dot{M}_S(t) = \phi \int_0^1 t(x,t)\ \mathrm{d}x - \nu M_S(t). \tag{41}$$

$\ell(x)$ integrates to 1, then

$$\dot{M}_U(t) = \nu - \nu M_U(t) - \phi\,\mathcal{T}(t), \qquad \dot{M}_S(t) = \phi\,\mathcal{T}(t) - \nu M_S(t), \quad \mathcal{T}(t) \equiv \int_0^1 t(x,t)\ \mathrm{d}x.$$

Adding (40) and (41) shows total population is constant:

$$\frac{d}{dt}\big(M_U(t) + M_S(t)\big) = \nu - \nu\big(M_U(t) + M_S(t)\big),$$

so if $M_U(T) + M_S(T) = 1$ then $M_U(t) + M_S(t) = 1$ for all $t \geq T$.

8.6 Initial condition at the switch

Let $(u_U^0(x), t^0(x), u_S^0(x), m_U^0(x), m_S^0(x))$ denote the pre-switch stationary densities under z_0. At the instant of the switch,

$$u_U(x, T^+) = u_U^0(x), \qquad t(x, T^+) = t^0(x), \qquad u_S(x, T^+) = u_S^0(x),$$

and segment masses satisfy

$$m_U(x, T^+) = m_U^0(x), \qquad m_S(x, T^+) = m_S^0(x).$$

For $t > T$, the system (35)–(39) evolves forward under the post-switch objects $(\tau, p^*, p_S^*, \theta_U, \theta_S)$ implied by regime z_1.